CAPSTONE PROJECT

AI SMART SCHEDULE & TASK PRIORITIZER

AN AI-POWERED STUDY COMPANION FOR STUDENTS

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PROBLEM STATEMENT

Students often juggle multiple assignments, exams, and extracurricular activities, leading to:

- Poor time management
- Procrastination and last-minute cramming
- Difficulty prioritizing tasks based on deadlines and personal focus patterns

Existing tools are generic, do not adapt to individual productivity, and lack intelligent, explainable scheduling.

PROPOSED SOLUTION

AI Smart Schedule & Task Prioritizer – a generative AI system that:

- Accepts tasks in **natural language**
- Parses deadlines, estimated effort, and priority using **Gemini AI**
- Learns each student's **best focus hours** from feedback (focus ratings 1-5)
- Generates an **optimized daily schedule** using a priority scoring algorithm
- Explains the schedule in **plain English** (AI coach)
- Sends **reminders** and adapts to delays

SYSTEM APPROACH

Tech stack used :

■ Language	Python 3.9+
■ AI / LLM	Google Gemini (google-genai SDK)
■ Scheduling	Custom greedy algorithm
■ Database	SQLite (local, no external setup)
■ UI / Interaction	Command Line Interface (CLI) + optional Telegram bot
■ Deployment	Streamlit / GitHub
■ Key libraries	google-genai, python-telegram-bot (optional), sqlite3, re, json, datetime

ALGORITHM & DEPLOYMENT

Priority Scoring Algorithm

- Score = 0.5×Urgency + 0.2×Effort + 0.3×UserHint + ProcrastinationRisk
- Urgency = 1 / (days until deadline + 0.5)
- Effort = normalized hours (log scale)
- UserHint = low(0.2), medium(0.6), high(1.0)
- ProcrastinationRisk = historical delay factor per task type

Scheduling (Greedy)

- Sort tasks by priority score (descending)
- For each day, get free blocks from user’s best focus hours
- Assign highest-priority task to earliest available block
- If a task doesn’t fit, push to next day and increase its score

Deployment

- Local: Run python smart_scheduler.py
- Planned: Web interface using Streamlit or Gradio
- Live demo: *Deployment link to be added after hosting*

RESULT

Working prototype with all core features implemented:

- Natural language task parsing (LLM + regex fallback)
- Automatic priority scoring and schedule generation
- Productivity learning via user feedback (focus ratings)
- AI-generated motivational explanations

Terminal output :

```
📅 Your AI-Generated Schedule:
  Wed 09:00 - study for physics exam (2026-06-04T12:00:00)
💡 Coach says: I placed physics prep at 9 AM because you focus best in the morning and it's due soo
```

CONCLUSION

- AI can effectively assist students in time management by automating **task prioritization** and **schedule creation**.
- The use of **generative AI (Gemini)** makes the system natural to interact with and explainable.
- **Personalised learning** (focus hour tracking) leads to better task placement and reduced procrastination.
- The project demonstrates a practical, low-cost solution using free APIs and lightweight storage.

FUTURE SCOPE

- **Google Calendar integration** (read fixed events / write schedule)
- **Full Telegram bot** with inline buttons and voice input
- **Mobile app** (Flutter) for on-the-go task entry
- **Collaborative scheduling** for study groups
- **Procrastination prediction** using a simple ML model (logistic regression) on historical logs
- **Deployment on Streamlit Cloud** for public access

REFERENCES

Google Gemini API – ai.google.dev

Streamlit • A faster way to build and share data apps

Welcome to Python.org

Supervised learning — scikit-learn 1.8.0 documentation

GITHUB LINK

- Github Link : https://github.com/AsmitaPokharkar/AI-Smart-Schedule-Task-Prioritizer
- Deployment Link: https://ai-smart-schedule-task-prioritizer-4rlisuowdoypayh5xdul4x.streamlit.app/

THANK YOU